


3.18

$$\widehat{\text{Insurance}} = 6.855 + 3.88 \text{ INCOME}$$

$$n=20, \text{截} = 7.383, \bar{X} = 59.3$$

$$\text{斜} = 0.112$$

a. $\text{Income} = 0 \Rightarrow \text{Insurance} = 6.855$

$$\text{Income} \uparrow 1 \Rightarrow \text{Insurance} \uparrow 3.88$$

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$$

$$\Rightarrow \bar{Y} = 6.855 + 3.88(59.3) = 236.939 \quad \#$$

b. $\hat{\beta}_1 = 3.88$

$$95\% \text{ CI: } \hat{\beta}_1 \pm t_{0.025, 18} \cdot se$$

$$\Rightarrow 3.88 \pm 2.101(0.112)$$

$$\pm 0.235$$

$$\Rightarrow \text{CI} = (3.645, 4.115) \quad \#$$

c. $\hat{Y} = 6.855 + 3.88(100) = 394.855$

$$\text{Var}(\hat{Y}) = \text{Var}(\hat{\beta}_0) + X^2 \text{Var}(\hat{\beta}_1) + 2X \text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$$

$$\text{Var}(\hat{\beta}_0) = 7.383^2 = 54.52$$

$$\text{Var}(\hat{\beta}_1) = 0.112^2 = 0.012544 \quad \Rightarrow \text{Var}(\hat{Y}) = 30.76$$

$$\text{Cov} = -0.746$$

$$X = 100$$

$$se = \sqrt{30.76} \approx 5.55$$

$$99\% \text{ CI: } t_{0.005, 18} \approx 2.878$$

$$394.855 \pm 2.878(5.55) = 394.855 \pm 15.97$$

$$\Rightarrow (378.89, 410.83) \quad \#$$

d. $H_0: \beta_1 = 5$

$H_1: \beta_1 \neq 5$

$$t = \frac{3.88 - 5}{0.112} = -10$$

Reject region: $|t| > 2.10$, $|-10| > 2.10$

$\therefore$ Reject H_0 $\#$

e. $H_0: \beta_1 = 5$

$H_1: \beta_1 > 5$

$$t = \frac{3.88 - 1}{0.112} = 25.71$$

Reject region: $t > t_{0.01, 18} \approx 2.552$

$\therefore 25.71 > 2.552$

$\therefore$ Reject H_0 保險增加速度大於收入